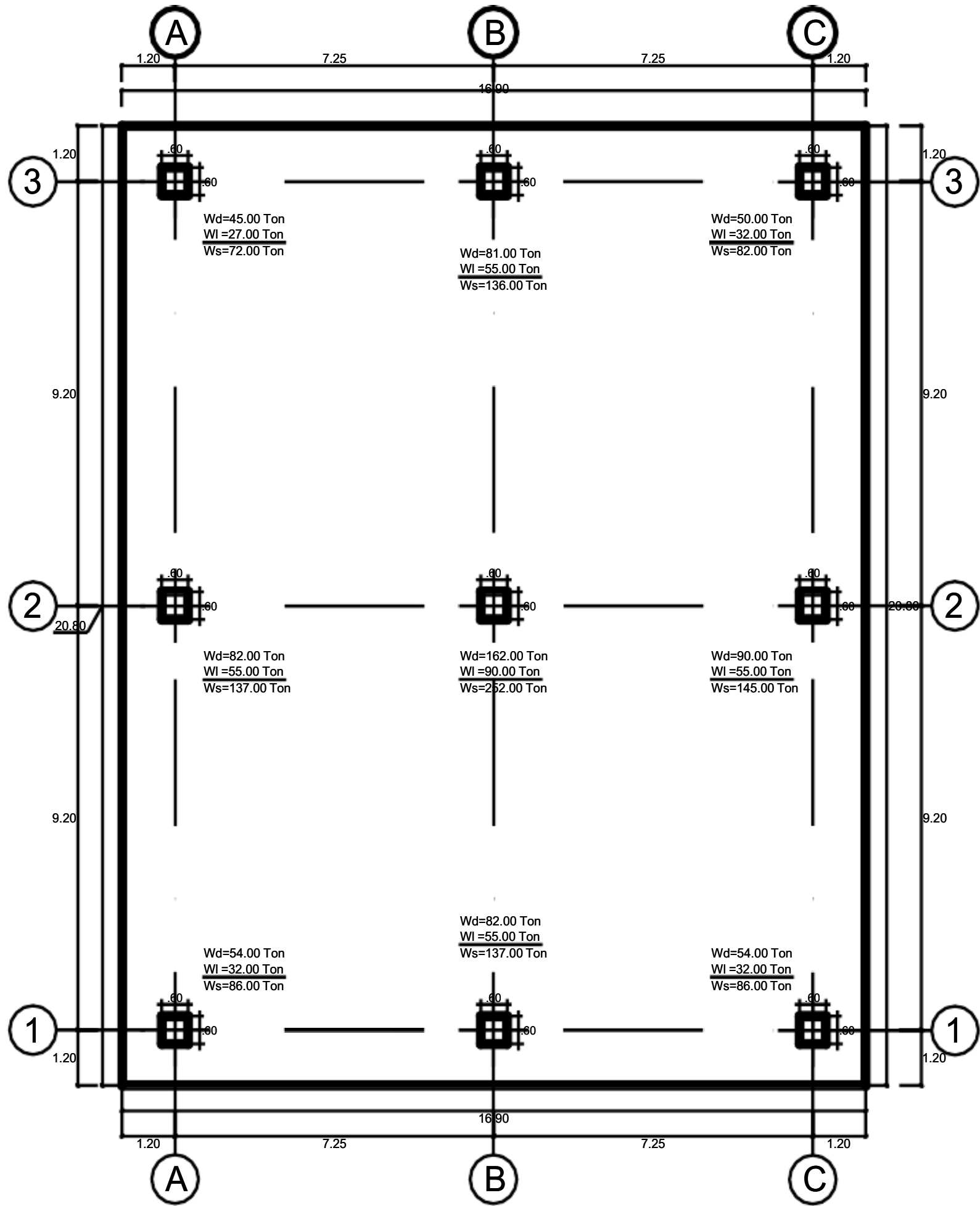
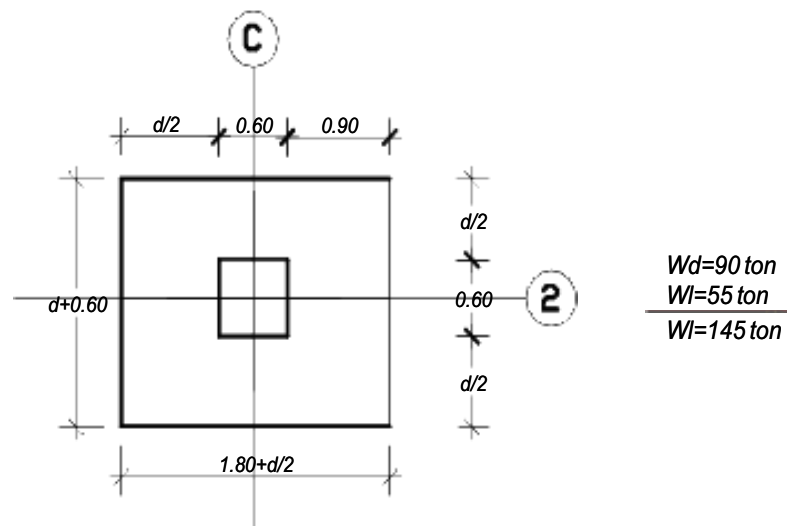


PROBLEMA Nº 01: PLATEA DE CIMENTACION

Analizar y diseñar la losa de cimentación con cargas en las columnas de sección transversal cuadrada de 60cm de lado, según estudio de mecánica de suelos EMS $t=0.90\text{ kg/cm}^2$, utilice para el diseño $f'c=210\text{ kg/cm}^2$, $f'y=4200\text{ kg/cm}^2$.



PLANTA - CIMENTACION

DIMENSIONAMIENTO ALTURA DE h Y d**Columna perimetral critico (C, 2)**

$$P_u = 1.7 W_l + 1.4 W_d \quad P_u = 219.50 \text{ Ton.}$$

Perímetro de punzonamiento

$$b_o = 2 \left(1.80 + \frac{d}{2} \right) + d + 0.60 \quad b_o = 4.20 + 2d$$

Condición de la Norma: $V_u < \phi V_c \rightarrow V_u = \phi V_c$ condición mínima

$$\phi V_c = \phi 0.53 \sqrt{f_c} * b_o * d \quad \phi V_c = 2741917 d + 13.057 d^2$$

$V_u = \phi V_c$ condición mínima

$$\phi V_c = 2741917 d + 13.057 d^2 = V_u = P_u \quad V_u = \phi V_c \text{ condición mínima } d = 61.842 \text{ cm.} \Rightarrow d = 60 \text{ cm.}$$

CARGAS DE SERVICIO:

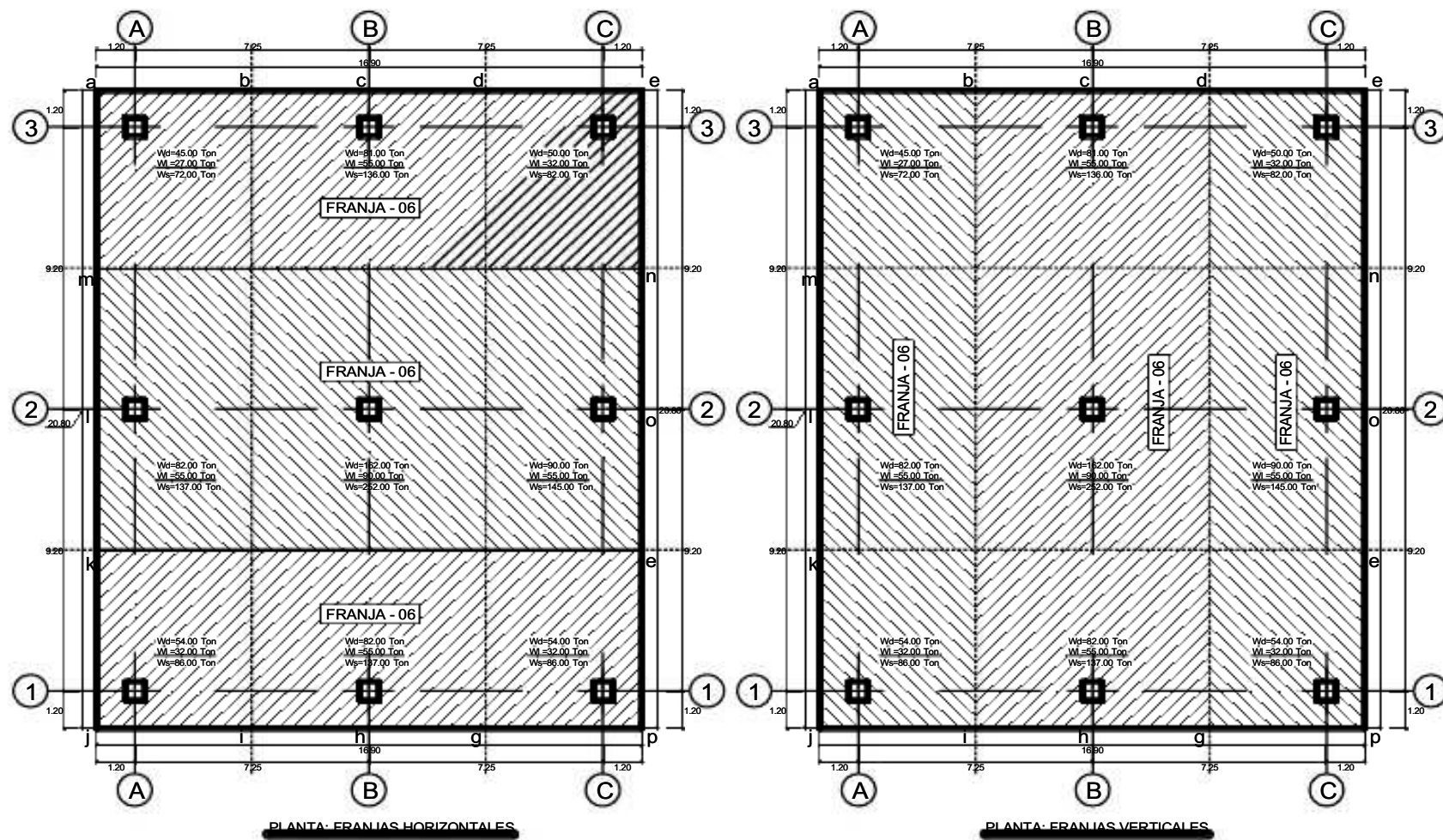
$$P_d = 45 + 81 + 50 + 82 + 162 + 90 + 54 + 82 + 54 \Rightarrow P_d = 700.00 \text{ Ton.}$$

$$P_l = 27 + 55 + 32 + 55 + 90 + 55 + 32 + 55 + 32 \Rightarrow P_l = 433.00 \text{ Ton.}$$

$$P_s = 1133.00 \text{ Ton.}$$

$$P_u = 1.7 W_l + 1.4 W_d$$

$$P_u = 1716.10 \text{ Ton.}$$



MOMENTOS DE INERCIA DE LA CIMENTACION:

Con respecto a "X": $I_x = \frac{16.90 \times 20.80^3}{12}$ $I_x = 12673.468 \text{ m}^4$

Con respecto a "Y": $I_y = \frac{20.80 \times 16.90^3}{12}$ $I_y = 8366.469 \text{ m}^4$

CENTRO DE GRAVEDAD DE LA CIMENTACION "Cg": (X,Y)

$$C_g = (16.90/2, 20.80/2) \Rightarrow$$

$$C_g = (7.25, 9.20)$$

CENTRO DE RIGIDEZ DE LA CIMENTACION "Cr": (X,Y)

$$\sum M_y' = 0$$

$$1133 X' = 7.25 (136 + 252 + 137) + 14.50 (82 + 145 + 86)$$

$$X' = 7.365 \text{ m.}$$

$$e_x = X' - X \quad e_x = 7.365 - 7.25 \quad e_x = 0.11518 \text{ m.}$$

$$\sum M_x' = 0$$

$$1133 Y' = 9.20 (137 + 252 + 145) + 18.40 (72 + 136 + 82)$$

$$Y' = 9.046 \text{ m.}$$

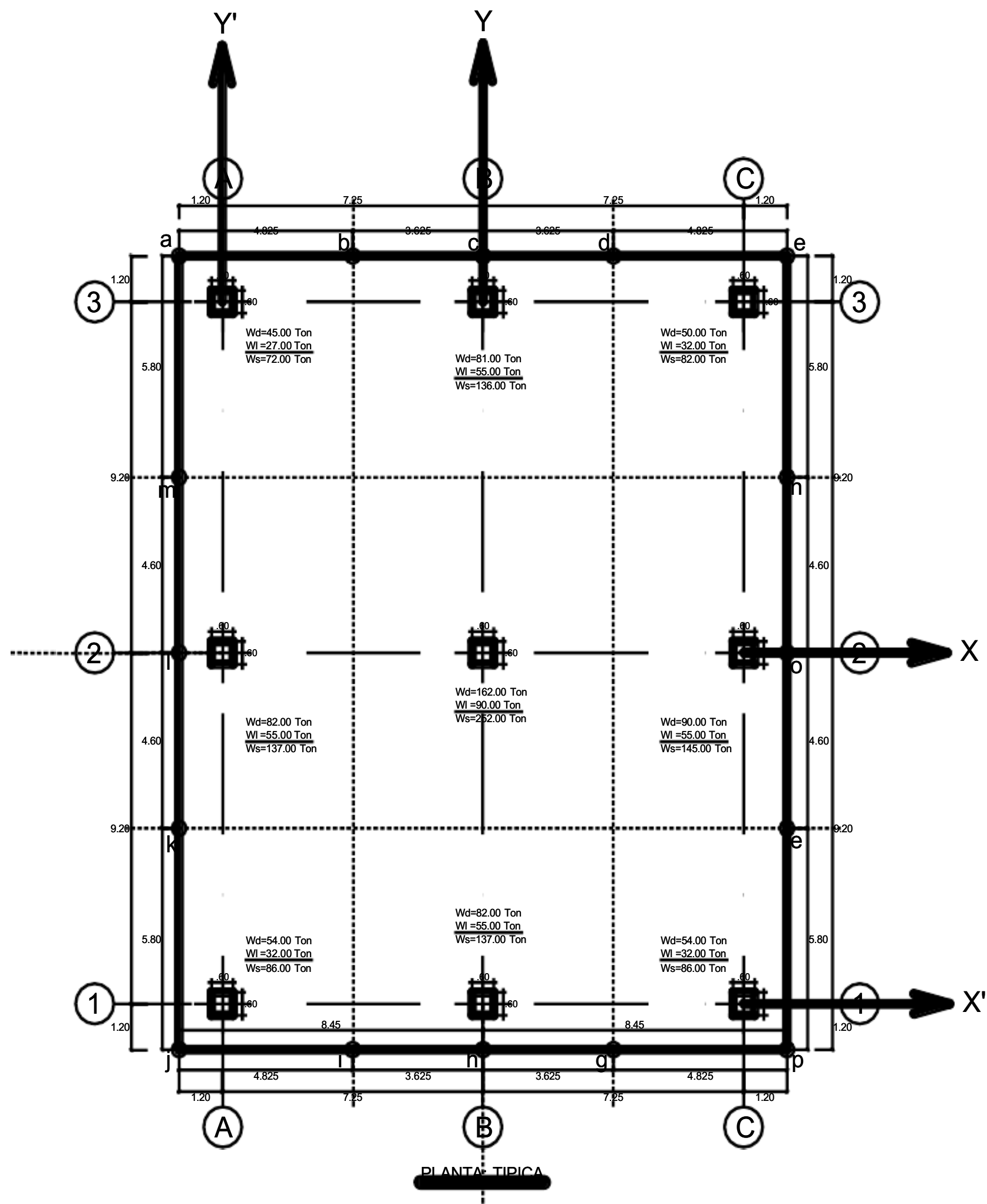
$$e_y = Y' - Y \quad e_y = 9.046 - 9.20 \quad e_y = -0.15428 \text{ m.}$$

$$C_r = (7.365, 9.046)$$

$$e = C_r - C_g$$

$$e = (0.115, -0.154)$$

$$C_g = (7.250, 9.200)$$



MOMENTOS CAUSADOS POR LA EXCENTRICIDAD:

$P_u=1716.10\text{ Ton.}\quad A=351.52\text{ m}^2.$

$E_y=-0.154\quad I_x=12673.468\text{ m}^4$

$E_x=0.115\quad I_y=8366.469\text{ m}^4$

Q

$\frac{M_y * X}{I_y} \pm \frac{M_x * Y}{I_x}$

$M_x = P_u * e_y\quad M_x = 1716.10 * -0.154\quad M_x = -264.279\text{ Ton} - \text{m}.$

$M_y = P_u * e_x\quad M_y = 1716.10 * 0.115\quad M_y = 197.352\text{ Ton} - \text{m}.$

$q = \frac{P_u}{A} \pm \frac{M_y X}{I_y} \pm \frac{M_x Y}{I_x}\quad q = \frac{1716.10}{9.65*20.80} \pm \frac{197.352}{8366.469} X \pm \frac{-264.279}{12673.468} Y$

$q = \frac{P_u}{A} \pm 0.02359 X \pm (-0.02085 Y)$

FRANJAS VERTICALES

		base y	0.02359	base x	0.02085 (ton/m2)	(kg/cm2)	
Puntos	Pu/A	X	0.02359 X	Y	0.02085 Y	q	q
a	4.882	-8.45	-0.199	10.4	-0.217	4.466	0.447
b	4.882	-3.625	-0.086	10.4	-0.217	4.580	0.458
c	4.882	0	0.000	10.4	-0.217	4.665	0.467
d	4.882	3.625	0.086	10.4	-0.217	4.751	0.475
e	4.882	8.45	0.199	10.4	-0.217	4.864	0.486
f	4.882	8.45	0.199	-10.4	0.217	5.298	0.530
g	4.882	3.625	0.086	-10.4	0.217	5.184	0.518
h	4.882	0	0.000	-10.4	0.217	5.099	0.510
i	4.882	-3.625	-0.086	-10.4	0.217	5.013	0.501
j	4.882	-8.45	-0.199	-10.4	0.217	4.899	0.490

Reacciones promedio en el suelo:

Reacciones por franja

Franja N°01: (a-b) y (i-j): (Ancho=4.825)

Reacciones franja 01 (a-b)(i-j):

$q_1=(q_a+q_b)/2=$

4.523

Ton/m2

$q_2=(q_i+q_j)/2=$

4.956

Ton/m2

21.822

Ton/m

23.914

Ton/m

4.825

l= 20.8

Area= A1=

100.36

m2

$(q_1+q_2)/2 * A1$

475.6578919

Ton

Franja N°02: (b-c-d) y (i-h-g): (Ancho=7.25)

Reacciones franja 01 (a-b)(i-j):

$q_1=(q_b+q_c+q_d)/3=$

4.665

Ton/m2

$q_2=(q_i+q_h+q_g)/3=$

5.099

Ton/m2

33.822

Ton/m

36.966

Ton/m

7.25

l= 20.8

Area= A1=

150.8

m2

$(q_1+q_2)/2 * A1$

736.1967456

Ton

Franja N°03: (d-e) y (g-f): (Ancho=4.825)

Reacciones franja 01 (a-b)(i-j):

$q_1=(q_d+q_e)/2=$

4.808

Ton/m2

$q_2=(q_g+q_f)/2=$

5.241

Ton/m2

23.196

Ton/m

25.289

Ton/m

4.825

l= 20.8

Area= A1=

100.36

m2

$(q_1+q_2)/2 * A1$

504.2453626

Ton

Reacciones por franja:

Suma de reacciones x franjas

Sum Reac.

1716.1

<>

Pu=

1716.10

FRANJAS HORIZONTALES

Puntos	Pu/A	base y		base x		0.02085 (ton/m ²) (kg/cm ²)	
		X	0.02359 X	Y	0.02085 Y	q	q
j	4.882	-8.45	-0.199	-10.4	0.217	4.899	0.490
k	4.882	-8.45	-0.199	-4.6	0.096	4.779	0.478
l	4.882	-8.45	-0.199	0	0.000	4.683	0.468
m	4.882	-8.45	-0.199	4.6	-0.096	4.587	0.459
a	4.882	-8.45	-0.199	10.4	-0.217	4.466	0.447
e	4.882	8.45	0.199	10.4	-0.217	4.864	0.486
n	4.882	8.45	0.199	4.6	-0.096	4.985	0.499
o	4.882	8.45	0.199	0	0.000	5.081	0.508
p	4.882	8.45	0.199	-4.6	0.096	5.177	0.518
f	4.882	8.45	0.199	-10.4	0.217	5.298	0.530

Reacciones promedio en el suelo:

Franja N°04: (j-k) y (p-f): (Ancho=5.80)

$$q_1 = (q_j + q_k) / 2 = 4.839 \text{ Ton/m}^2$$

$$q_2 = (q_p + q_f) / 2 = 5.238 \text{ Ton/m}^2$$

$$30.378 \text{ Ton/m}$$

Reacciones por franja

Reacciones franja 04 (j-k)(p-f):

$$a = 5.80 \quad l = 16.90$$

$$\text{Area} = A_1 = 98.02 \text{ m}^2$$

$$(q_1 + q_2) / 2 * A_1 = 493.8557621 \text{ Ton}$$

Franja N°05: (k-l-m) y (n-o-p): (Ancho=9.20)

$$q_1 = (q_k + q_l + q_m) / 3 = 4.683 \text{ Ton/m}^2$$

$$43.080 \text{ Ton/m}$$

$$q_2 = (q_n + q_o + q_p) / 3 = 5.081 \text{ Ton/m}^2$$

$$46.748 \text{ Ton/m}$$

Reacciones franja 05 (k-l-m)(n-o-p):

$$a = 9.20 \quad l = 16.90$$

$$\text{Area} = A_1 = 155.48 \text{ m}^2$$

$$(q_1 + q_2) / 2 * A_1 = 759.0442308 \text{ Ton}$$

Franja N°06: (j-k) y (p-f): (Ancho=5.80)

$$q_1 = (q_d + q_e) / 2 = 4.526 \text{ Ton/m}^2$$

$$26.252 \text{ Ton/m}$$

$$q_2 = (q_p + q_f) / 2 = 4.925 \text{ Ton/m}^2$$

$$28.564 \text{ Ton/m}$$

Reacciones franja 06 (j-k)(p-f):

$$a = 5.80 \quad l = 16.90$$

$$\text{Area} = A_1 = 98.02 \text{ m}^2$$

$$(q_1 + q_2) / 2 * A_1 = 463.2000071 \text{ Ton}$$

Reacciones por franja:

Suma de reacciones x franjas

Sum Reac.

1716.10

<>

Pu=

1716.10

